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BY

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**Autonomous Visual Docking of a ROV onto a
Moving Dock: Terminal Guidance in Simulation
and Empirical Limits of ArUco Perception
Underwater**

Thesis submitted as a requirement for the degree of Bachelor of
Engineering

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ORIGINALITY STATEMENT

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Abstract

This thesis focuses on the optical terminal stage of underwater docking using passive markers and computer vision, the approach available to low-cost ROV platforms that cannot carry learning-based perception. In simulation, a position-based visual servoing pipeline docks a BlueROV2 autonomously onto a swaying dock through two phases: coarse approach and fine alignment. In parallel, the operating envelope of passive ArUco perception is measured from real underwater imagery through analysis of detection range, pose error, and degradation with distance and turbidity. Together, these bound the operating range of marker-based terminal docking and identify where a paired complementary far-field cue becomes necessary.

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1 Introduction

Docking provides an underwater vehicle a place to recharge, offload its data and perform long-running tasks without surfacing or returning to a support ship. Since surfacing and recovery are slow, and weather-limited and expensive, removing them from the mission cycle is what makes long deployments and resident vehicle systems possible. The engineering challenge concentrates in the last few metres of the dock approach (Figure 1), in which if this stage goes wrong, the vehicle may collide with the dock, or abort and spend unnecessary time and energy retrying.

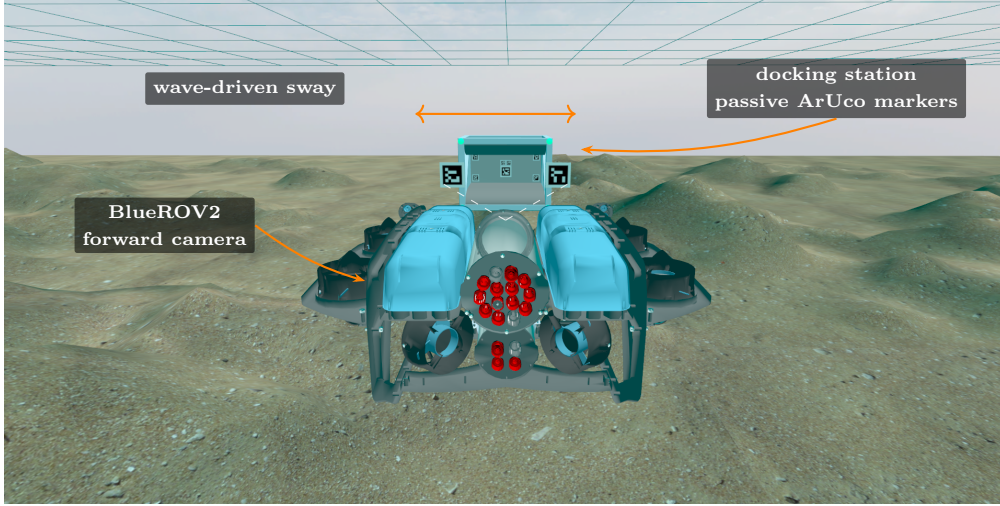


Figure 1: The terminal docking problem: the vehicle must fly the final metres onto a docking station using its forward camera and marker detection, while the dock itself sways with the sea above.

Conventionally, a docking mission is staged by sensing modality, like acoustic or electromagnetic homing, to bring the vehicle from long range into the vicinity of the dock, then vision takes over for the final metres, where the precision requirement is tightest. This thesis addresses only the vision stage; homing and the mechanical capture are out of scope. Within the vision stage, two practical difficulties receive little attention in the literature. First, a real dock is rarely still, since moored and suspended structures inherit the wave motion of the sea around them. Second, the water limits how far a camera can see the dock at all, so marker visibility itself becomes a design constraint.

The work is split into two parts (Figure 2). Part A develops the complete docking pipeline in simulation (marker fusion, a Kalman filter for the dock pose, and phase control) and runs it against a dock swaying at certain wave periods.¹ A docking trial sweep through the sway periods is performed to determine where the docking breaks down and the failure analysis explains why. Part B takes the marker detector into real water where a pool capture measures detection against marker size, maximum detection range and the water’s attenuation coefficient, and the same frames are then processed with artificial turbidity to produce a marker sizing rule. This split is necessary since the Gazebo simulator is able to exercise the closed loop, but cannot render water optics; the pool test captures have real optics but no control loop.

¹The docking stack and simulation fork are available at <https://github.com/alanchoi00/bluerov2-docking> and <https://github.com/alanchoi00/blue-sim>.

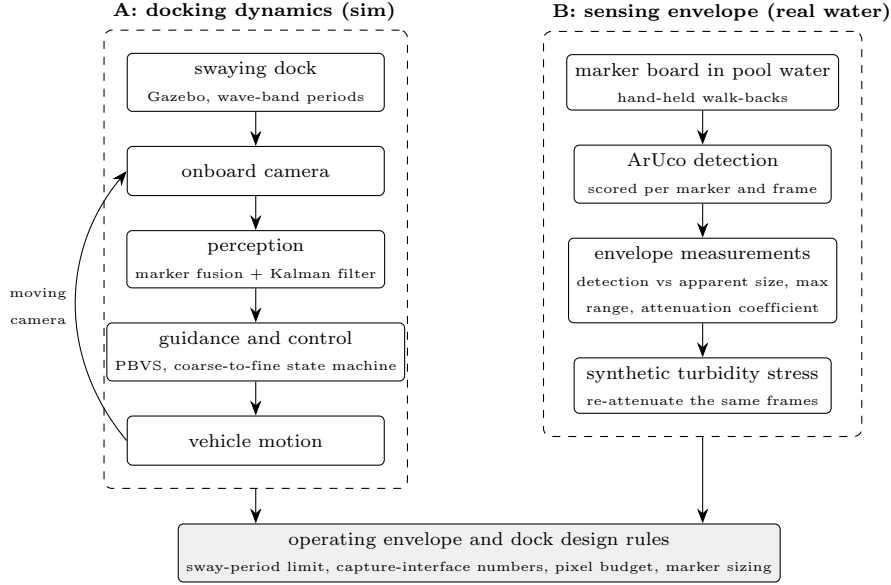


Figure 2: How the two parts fit together. The return arrow in Part A is the closed control loop; Part B is an open-loop measurement chain; their outputs combine into the envelope and design rules.

The objective of this thesis is to build and validate a terminal docking stage from passive markers and computer vision, the approach available to a low-cost ROV, and to establish its operating envelope. In simulation, the system docks autonomously onto a swaying dock, cleanly down to 8 s sway periods at 0.1 m amplitude. The swept periods are mapped to the NSW sea states to determine suitable dock depths. The sweep also uncovered a control instability: estimating the dock’s velocity through a moving camera lets the vehicle’s own motion corrupt the estimate, and at short sway periods this feedback destabilises the approach, while a simpler design that never estimates dock velocity is unaffected. On the sensing side, measurements on real underwater imagery established the perception envelope: a marker needs around 27 pixels of apparent size to be detected, maximum detection range is about 35 times the marker side, the water’s attenuation coefficient was measured from the imagery itself, and inverting these numbers yields a sizing rule for the dock’s markers.

Section 2 reviews the docking literature and states the research gap. Section 3 describes the system and the experimental method for both parts. Section 4 presents and discusses the results, and Section 5 concludes with future work.

2 Literature Review

2.1 Underwater docking: motivation and phases

Autonomous Underwater Vehicle (AUV) docking has been a focus of research for decades, driven by the need for long-duration missions via subsea battery recharging and data transfer. A wide range of docking guidance strategies has been explored, as summarized in recent surveys [38]. Broadly, approaches are categorised by the sensing and guidance modality. Acoustic homing using sonar beacons can guide AUVs from long range, often providing range and bearing to a dock [19]. Electromagnetic

docking systems have also been tested (e.g. using low-frequency EM fields for short-range homing), notably in Feezor’s EM homing system [7]. Vision-based techniques, using cameras to detect visual markers, offer high precision in final alignment. In practice, multi-modal solutions are common: for example, an AUV might navigate toward a dock via acoustic homing signals and then switch to vision-based guidance in the final meters [19]. Dedicated docking markers can be active (such as flashing lights or LED beacons) to increase visibility, or passive high-contrast patterns that require no power, as demonstrated in Maire’s pole design. On the mechanical side, most docking station designs incorporate funnel-shaped guide surfaces to help capture the vehicle despite some misalignment. This combination of sensor guidance and mechanical alignment ensures a robust docking process under various conditions.

Numerous projects have demonstrated AUV docking with different infrastructures and homing strategies. One of the earliest was the WHOI REMUS AUV docking system from the late 1990s, which used a Relative Acoustic Tracking system of acoustic transponders for homing into a funnel-like subsea dock (see Figure 3) [33, 2].

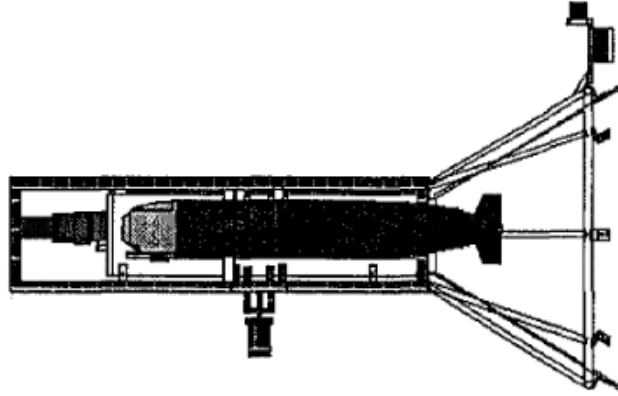


Figure 3: A side-view drawing of the WHOI REMUS AUV docking system [33].

Researchers have since tested seabed-mounted and moored docking stations to support AUV missions. For instance, Maire et al. (2009) proposed a heavy weighted docking target pole lowered to the seafloor, equipped with an acoustic beacon for long-range localisation and a visual pattern for final approach. An AUV can home to such a seafloor station, attach itself, and enter a sleep or recharge mode until the next mission window [19].

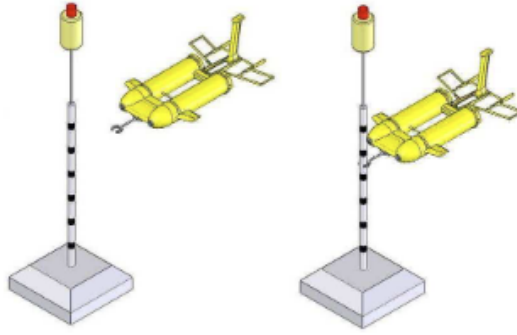


Figure 4: Maire et al.’s seabed docking station concept with acoustic and visual homing aids. (left: on approach; right: after docking) [19]

Remotely operated vehicles (ROVs), being tethered and human piloted, traditionally dock via their tether management system (TMS), a cage lowered from the surface into which a pilot must manually fly the vehicle. Trslic et al. demonstrated the first autonomous TMS docking of a heavy work-class ROV in open ocean conditions, using vision-based pose estimation of the four active light markers on a conventional TMS cage (Figure 5). The trials demonstrated repeated autonomous dockings to the suspended, heaving cage in January North Atlantic sea conditions [35].

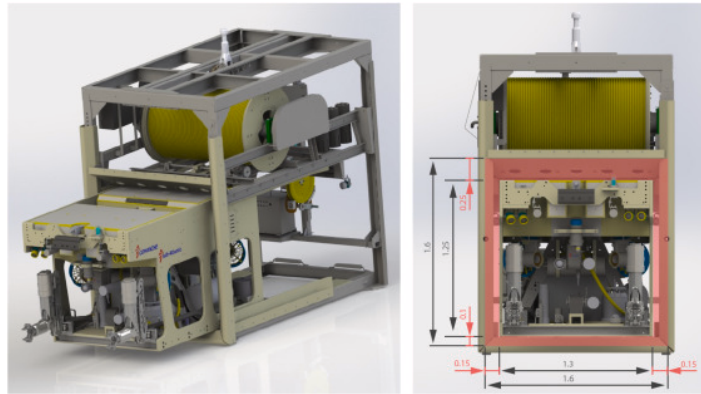


Figure 5: The tether management system docking cage for work-class ROVs used in the first autonomous TMS docking demonstration [35].

In a different configuration, researchers have also developed hybrid docking setups where an ROV acts as a mobile docking station to service AUVs. Morinaga et al.’s docking station ROV captures an AUV using electromagnets, and a slider device then aligns the receiving and transmitting antennas for inductive power transfer (see Figure 6). The concept was demonstrated in a water tank docking test [22].

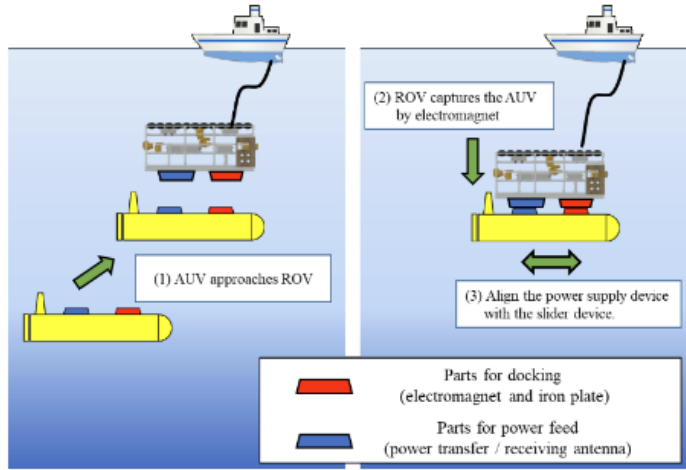


Figure 6: A flowchart of ROV-AUV docking interaction from Morinaga et al. [22]

This illustrates the breadth of ROV-involved docking solutions, from launch and recovery systems to ROVs serving as docking stations for AUV support. Across these systems, the common structure is a two-phase mission: long-range homing followed by short-range terminal alignment. This thesis addresses the terminal phase.

2.2 Acoustic and electromagnetic guidance

Acoustic-based navigation is widely used as the long-range guidance channel for underwater docking. A common configuration has an acoustic transceiver on the dock that emits a signal, which is received by a transponder on the AUV. The AUV then replies with its own signal, allowing the dock to compute the distance based on the time delay. To obtain bearing information, several inverted ultra-short baseline (iUSBL) receivers are spaced precisely on the AUV hull. By measuring the slight time difference of arrival of the acoustic signals at each receiver, the system can triangulate the relative azimuth and elevation angles between DS and AUV. WHOI's REMUS AUV docking system is a prime example of a USBL-navigation aided system [2]. Its acoustic homing can guide the AUV from long range, at the cost of moderate update rates, latency due to the speed of sound underwater, risk of false signals from acoustic refraction and ambient noise, and limited angular resolution.

Electromagnetic (EM) guidance offers an alternative to acoustic signals, using low-frequency magnetic fields emitted by a coil on the dock, and results in lower latency. Feezor et al. (2001) developed an EM homing system where a low-frequency magnetic field is generated by a coil on the dock, and the AUV uses magnetometers to detect the field strength and direction, without line-of-sight constraints [7]. Their experiments demonstrated that the AUV could home in on the dock from a limited range up to ~ 30 m. Further, the system requires dedicated coils, drivers and calibration of the local field environment, which may limit its practicality for general-purpose docking stations.

Most underwater docking stations incorporate mechanical alignment aids to relax the sensing requirement and accommodate for the slight tolerances during the final docking alignment phase. There exist two main configurations: unidirectional and omnidirectional guides. Unidirectional stations are

the most common type of configuration, typically designed for a torpedo-shaped vehicle approaching on a known heading. Omnidirectional stations allow approach from any heading, typically as a vertical pole or tensioned cable, allowing the vehicle to be latched onto the dock despite its orientation, notably in the Odyssey IIb AUV’s dock system [31], simplifying the navigation and alignment process at the cost of a dedicated latch mechanism on the vehicle nose. In either configuration, the guide’s capture radius and tolerated closing speed define the interface between the guidance problem and the mechanical one, in which the capture envelope assumed in this work (Section 3.2.5) plays that role.

2.3 Dock motion in ROV operations

During operational practice, the docking target is rarely still in place. For instance, Trslic’s docking system has a work-class ROV recovered into a TMS cage suspended from its support vessel, and the cage inherits the vessel’s wave-driven motion through the lift wire. Manual TMS docking requires the pilot’s full concentration and skill, flying into a cage whose heave, yaw and pendulum swing follow the sea state, while decisions are taken in fractions of a second [35]. The first autonomous TMS docking trials were run in dynamic North Atlantic wave conditions and were motivated by exactly this difficulty, with the objective to widen the operational weather window that the task currently dictates [35]. In some concepts, the dock is not even a fixed structure: a docking-station ROV captures and recharges an AUV in the water column, so both the dock and the AUV are free bodies in the system [22]. Even purpose-built resident docking stations are not still: they are typically moored or suspended structures rather than rigid foundations, and dynamic analysis of a seabed-moored suspended dock shows it responds measurably to ocean-current forcing transmitted through its mooring line [10].

This phenomenon of forcing on underwater object’s motion is ordinary ocean physics. Ocean surface waves carry their energy at period of roughly 5 to 20 s (crest-to-crest times). Waves raised by the local wind sit at the short end of this band (5 to 8 s period), while swell arriving from distant storms has longer periods (above 8 s) [11]. Below the surface, the wave orbital motion decays exponentially with depth at a rate set by the wavelength, while wavelength grows with the square of the period, so short wind sea is filtered out within tens of metres of depth while long-period swell induces wave motion far deeper [11]. A moored dock, therefore, oscillates with a period-dependent amplitude. In a typical sea, the motion surviving at continental-shelf depths is still of order 0.1 m, and because decay depends on period, a dock’s depth determines which wave periods reach it. Off the NSW coast, the operating context of this work, the long-term mean sea state is a wave height of 1.6 m at a mean period of 8 s [30, 16], which places realistic dock motion within the wave band swept in Part A. The quantitative mapping from period and depth to dock displacement is developed with the experimental design in Section 3.2.1.

Against this operational picture, the docking control literature almost always models the dock as a fixed pose in the world frame, and where environmental disturbance appears, it is a steady current applied to the vehicle. The exceptions are partial. Trslic et al. docked to a heaving TMS cage by servoing on the most recent measured pose, reacting to the motion without having to model it [35]. The same group later showed that TMS heave can be predicted (about 2.5 s ahead with a neuro-fuzzy model), but the prediction came from a pressure sensor mounted on the TMS cage itself, which only covers heave and was evaluated offline instead of in closed-loop docking [34]. Neither line of work

estimates the dock’s motion through the vehicle’s own camera while both bodies are moving. The estimation problem that configuration raises is taken up in Section 2.6.

2.4 Vision-based docking and fiducial markers

Passive visual markers are high-contrast patterns (typically black and white pixelated squared markers) detected by a camera and used for pose estimation. Early work from Maire’s Starbug dock system designed custom geometries which used a binary encoded pole as a passive black and white target mounted inside a suspended docking station. The pattern is designed so that it is visible from any arbitrary yaw, roll or pitch angle and also serves as a structural station that the AUV can grip on. Detection is performed with a fast Haar feature template matcher over the image, where the ‘dash-dot-dash’ black and white pattern is scored over multiple scales, positions and image rotations to locate the camera pose relative to the dock [19]. As the AUV approaches closer, the system switches to shorter templates to maintain more precise tracking at close distance. This method purely using a high contrast pattern can deliver robust homing without any powered beacons, however, it is limited by the CPU cost of the template search over multiple scales and rotations, and restricts design flexibility for the dock station. Later work on general visual application for regular (standardised) fiducial systems like AprilTag and ArUco follows the standard method of obtaining pose of the markers from camera frame, which square markers are detected by adaptive thresholding and decoded against a dictionary, with the four corners feeding a Perspective-n-Point (PnP) solution for full 6-DOF pose [8, 14]. Notably, Xu’s AUV navigation system detects a grid of multiple markers for improved accuracy [37]. The ArUco library first decodes each marker ID and recovers the marker’s pose relative to the camera. Then, Xu’s custom noise model computes an optimal combined pose estimate from all visible markers, improving accuracy and robustness against individual detection errors. Vision positioning with two cameras has likewise been used in AUV docking experiments [17]. The same principle applies to Romero-Ramírez’s fractal markers, which survive across the long to short range span by having “an aggregation of squared markers, one into another, in a recursive manner”, see Figure 7 [28].

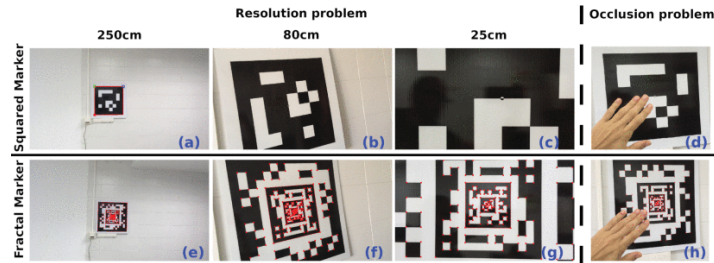


Figure 7: Result of the Fractal Markers against regular squared markers at different distances and occlusion: Fractal Markers can be detected in more cases than regular squared markers [28].

There is little design guidance on how these marker sets should be arranged. Wang’s controlled study on marker size, count and distribution found that apparent size is the dominant factor for localisation accuracy, while marker count and placement only have secondary effects [36]. The dock layout of this work assumes the same ordering. The closest deployed system to this work is Günzel’s resident mini-ROV, which docks onto a multi-size ArUco layout at 90 m depth, fused through an extended Kalman

filter, and reports a 90% docking success rate [9]. This architecture matches the size-graded layout and filter design of this work. However, the dock station is static, and the marker sizes were selected through simulator experiments without water optics, so the underwater perception envelope remains unmeasured. For underwater detection performance, dos Santos Cesar et al. evaluated three fiducial systems (ArUco, AprilTag and ARToolKit) in tank conditions, measuring the smallest detectable size, maximum viewing angle and detection range across six turbidity levels [6]. This study establishes apparent size and turbidity as the two main variables of underwater fiducial detection. Among the three libraries, ArUco showed the highest sensitivity to turbidity, with its required marker size growing at the fastest rate.

Most recently, Ni et al. proposed a hybrid active and passive docking marker where a printed AprilTag is placed at the center of the dock station, surrounded by an array of five LED lights [23], following Trslic’s four light beacon active marker design [35]. A custom YOLO-D model is trained to simultaneously detect the LED array and the AprilTag in raw underwater images.

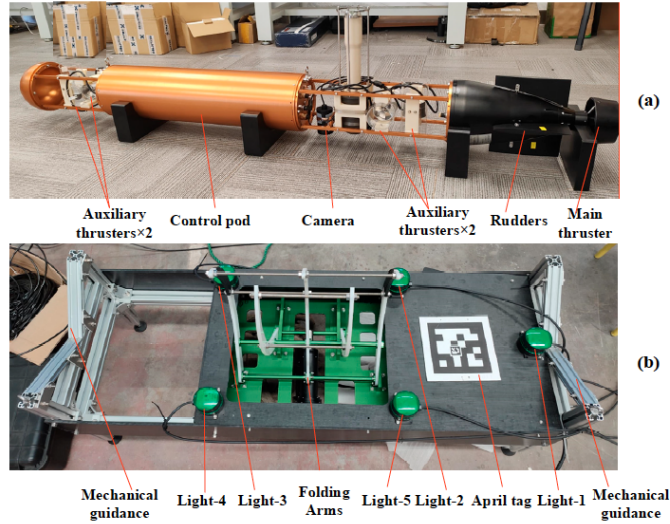


Figure 8: Example of an underwater docking station and vehicle: (a) AUV; (b) Docking station with five green LED beacons (Light-1 to Light-5), AprilTag marker at the center and v-shaped mechanical guides.[23]

Ni et al.’s AUV docking guidance is composed of two levels of detection:

- Level I: the LED array is used as visual markers for pose estimation. However, at close range, the light scattering can introduce errors in estimating the center of the guidance light source.
- Level II: the AprilTag is used for precise pose estimation. However, this guidance method is limited by the detection range and visibility of the tag under turbidity, water quality and illumination.

The docking process begins by capturing an image, which is preprocessed through configured filters to minimise noise in the image. The YOLO-D model then detects the center X_L of the LED array and the AprilTag region of interest (ROI) in the image frame. In Level I guidance, the obtained X_L is

used to compute a coarse position P_L of the docking station relative to the AUV. In Level II guidance, the AprilTag corners X_A are extracted from the ROI and used to compute a precise pose P_A via PnP. Finally, the pose estimates from both levels are fused to compute the overall P_{out} , following the formula:

$$P_{out} = K_1 P_L + K_2 P_A$$

where K_1 and K_2 are weight matrices adjusted according to the conditions, shifting toward the surviving channel when poor water quality prevents one marker type from being detected. This fusion allows the system to leverage the long-range detection of the LED array while maintaining the precision of the AprilTag when visible.

Learning-based detection is the other active research direction. Standard object detectors are typically trained on terrestrial images, so their performance degrades in low-light and low-contrast conditions [5], which are the typical conditions of underwater scenes, and motivated a few underwater-specific variants. For example, Pan’s YOLOv9s-UI adds attention modules for underwater object detection [25], and Ni’s YOLO-D (described above) augments a YOLO backbone with multi-scale feature fusion and adaptive kernels to detect distant lights and partially visible tags [23]. Another, Sans-Muntadas trained a CNN that regresses the heading error to the dock directly from raw images, replacing explicit marker detection entirely [29]. However, this method degrades at close range where visible features vanish. In practice, these learning methods serve as the detection front end of a larger system, while the metric pose is still recovered geometrically from the marker corners. They also carry extra costs in training data, onboard compute and validation effort, which are significant for small / low-cost ROV platforms. For this reason, this thesis stays with the classical fiducial pipeline. The detection algorithm is mature and cheap, and the open question is the operating envelope of the detector rather than the detector itself.

2.5 Visual servoing and guidance for docking

Visual servo control divides into two schemes. Image-based visual servoing (IBVS) regulates the feature error directly in the image plane, while position-based visual servoing (PBVS) reconstructs a metric relative pose and regulates it in Cartesian space [4]. In this work, the perception layer already produces a full 6-DOF pose from fiducial markers of known geometry using the standardised marker PnP algorithm [8]. Therefore, position-based servoing adds no extra cost, and it keeps the docking geometry expressible in physical units, which is why it is adopted here.

However, the servoing literature assumes that the target pose is fixed. For the moving docks of Section 2.3, the servo target itself oscillates, and regulating onto it raises the estimation question taken up next.

2.6 State estimation for a moving dock

When the dock moves, the guidance layer of the control system needs a model of the dock’s motion in addition to its pose. The target-tracking literature supplies a standard kinematic hierarchy for this [3]. A constant-position model treats the target as static and absorbs all dock relative motion as

process noise on the position. A nearly-constant-velocity model augments an additional velocity state driven by white acceleration noise, and manoeuvre models add further derivatives or switching. The acceleration noise density is the main tuning parameter as it sets the filter’s bandwidth, which sets the filter’s bandwidth as a trade-off between responsiveness to real target motion and amplification of measurement noise. In regards to this thesis, two properties of this hierarchy matter. First, the constant-position model is a different estimator since it does not have a velocity state and cannot extrapolate, which limits its tracking against a moving target, but immune to velocity estimation corruption. Second, every member of the hierarchy inherits the Kalman assumption that measurement errors are white noises and independent of the state. The filter offers no guard against this false assumption and no process-noise setting can retune it away [3]. The two filter properties are compared in Part A and its failure analysis is a case study of the second property.

This kind of filter now appears routinely inside docking systems. For example, Günzel’s resident-vehicle system fuses ArUco pose with acoustic positioning via an extended Kalman filter (eKf) [9], and Ni’s system fuses two visual pose changes by confidence weighting [23]. In both systems however, the dock is static, and the filter suppresses noise and outliers rather than estimating target motion. As a result, tracking a moving dock from a moving vehicle raises further problems. The marker measurement is reconstructed in the world frame through the vehicle’s own estimated pose, so vehicle motion and localisation error leaks into the measured dock position. Therefore, this measurement error becomes correlated with the observer’s state. This correlated measurement noise is a recognised failure mode in target-tracking literature, with known flaws such as augmenting the filter with the observer’s state or estimating in relative frame [3]. In the underwater docking literature, however, this failure mode exposes an unexamined gap. Against a static dock, the leakage is harmless and absorbed as position noise, whereas a velocity state integrates it into a target velocity. Part A demonstrates and analyses this mechanism, including the case where the corrupted velocity estimate feeds a control loop.

2.7 Underwater image formation, turbidity and simulation

The light attenuation physics of underwater imaging is described by the Jaffe–McGlamery model, which is composed of three components at the sensor: the direct light attenuated along the line of sight, the forward-scattered light that blurs the direct image, and the backscattered veiling light that adds a range dependent haze [20, 13]. Most applications use a simplified form that keeps only the attenuation and veiling components, $I = Je^{-\beta d} + B(1 - e^{-\beta d})$. This simplification though has known limits. Akkaynak and Treibitz showed that the standard form mixes up two physically distinct attenuation coefficients, with consequences for colour reconstruction [1]. Further, Ismirohlou et al.’s recent tank experiments put the forward-scatter (blur) term back into the simplified model. Their results show that in highly turbid water, images blur even at short distance to the camera [12]. Part B of this work adopts the simplified model, measures its β from the target itself in the frame, then tests the blur question at the optical depths reached.

Marker detection under turbidity has also been approached with learning methods. Risholm et al. trained a marker detector with predictive uncertainty, which resulted in 100% detection rate across attenuation lengths where the standard ArUco library’s detection rate collapses, with the median translation error growing from 2.6 cm at low turbidity to 10.5 cm in higher turbidity [26]. DeepTurbid’s

approach replaces the four-corner detection with 68 learning keypoints per marker and trains on synthetically degraded imagery. It reports 95.8% recall against OpenCV’s 9.9% on its turbid test set [21]. Both results show that learning buys detection recall in murky water at a cost in compute and localisation precision, so the classical (PnP) pipeline remains the right baseline for this thesis. Notably, DeepTurbid’s training data is generated by re-attenuating clear in-air imagery through the simplified image formation model. This procedure is used in Part B, by degrading real underwater frames with geometric range truth and a measured attenuation coefficient.

Simulation platform occupy the other end of the fidelity spectrum. The standard robotics simulator like Gazebo model rigid-body dynamics, buoyancy and hydrodynamic drag well enough to test the full control stack. However, its renderer model neither light attenuation nor veiling light, so simulated marker detection is optimistic. Physically grounded underwater image simulation exists such as OceanSim, but it does not accurately model vehicle and fluid dynamics, while Gazebo model the dynamics but not the optics. Currently, no open-sourced robotics platform provides both halves with validated fidelity, so this division motivates the two-part structure of this thesis.

2.8 Research gap

This thesis contributes to three recurring limitations. First, in existing literatures, the dock is almost always assumed static, which reduces the control problem to steering into a fixed pose, with the exception of handling dock motion with dock mounted sensors. No work estimates a dock’s wave-frequency motion onboard, through vision alone, from a vehicle that is itself moving and has a close loop control on that estimate. Second, vision-based docking studies are usually evaluated in clear tanks and pools, where turbidity and illumination are controlled or only coarsely varied [18, 23]. The quantitative envelopes across turbidity, range and marker size are largely missing, with Cesar et al.’s tank evaluation of dos Santos as an exception. Third, the learning-based systems that report the strongest underwater robustness assume GPU-class onboard compute and has training sets that are poorly matched to low-cost ROV platforms with limited power and compute.

This thesis addresses these gaps in two parts. Part A quantifies the dynamic envelope, which has an end-to-end docking of a BlueROV2 against a dock swaying at wave periods while mapping to real sea states, with the full dynamic marker perception pipeline within the loop, including the failure mode of introducing velocity estimation derived from a moving camera. Part B quantifies the sensing envelop with real underwater imagery, providing a pixel budget for ArUco detection, a measured light attenuation coefficient and a marker-sizing rule derived by adding high turbidity to real frames. Together, these two parts bound where a passive-marker docking system of this class can operate, a characterisation the literatures currently lacks.

3 Methodology

3.1 System architecture

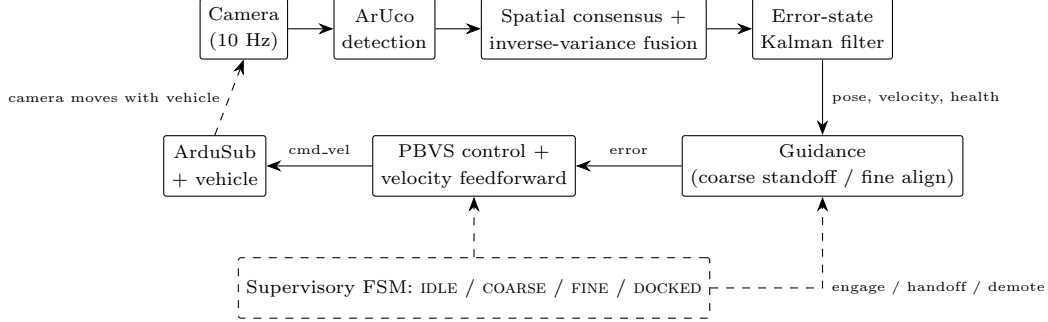


Figure 9: System architecture. The state machine gates the controllers on the filter’s health signal; the vehicle-to-camera feedback path is the ego-motion coupling analysed in Section 4.1.

The system includes a perception, guidance and control sub-pipeline for the terminal stage of docking. Perception estimates the dock pose from the ArUco markers on the dock and publishes a filtered pose, a dock velocity estimate, and a semantic health signal. Guidance converts the dock pose and the vehicle pose into a body frame error to a target. Control regulates that error with position-based visual servoing (PBVS) and writes a body velocity command. A supervisory state machine sequences the mission through 4 states (with demotion paths from loss of perception):

1. IDLE: vehicle not active or on hold.
2. COARSE: vehicle drive to a standoff point on the dock entry axis.
3. FINE: vehicle fine aligns with the dock centre pose and advance into entry.

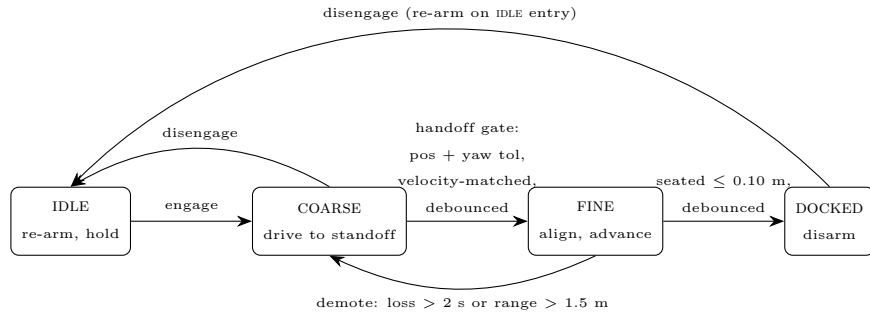


Figure 10: Docking state machine with the transition gates annotated.

Position based servo-ing was selected over image-based servo-ing because the marker pipeline already produces a full 6-DOF pose by fusing multiple markers of known geometry and relative positions yields a calibrated dock pose, so servo-ing in pose space costs nothing extra.

The scope boundary is the optical terminal stage, meaning the trails begin with the dock inside the camera field of view at several metres range. Long-range acquisition (acoustic homing or an active light cue) is assumed to have guided the vehicle closer to this point and is out of scope.

3.2 Part A: Docking simulation

3.2.1 Simulation environment

The simulation runs on ROS 2 Jazzy with Gazebo (gz-sim 8, DART physics) and a battle-tested ArduSub autopilot with ArduPilot Gazebo plugin. The BlueROV2 Heavy vehicle model, its hydrodynamic and buoyancy plugins, and the simulation bring-up derive from the open-source Blue framework [27, 24], of which this project maintains a git fork adapted to ROS 2 Jazzy. The simulated BlueROV2 has a forward camera at 10 Hz. The world places a docking station 5 m ahead of the vehicle spawn at 1.5 m depth, with all nine ArUco markers (47.5 mm to 200 mm).

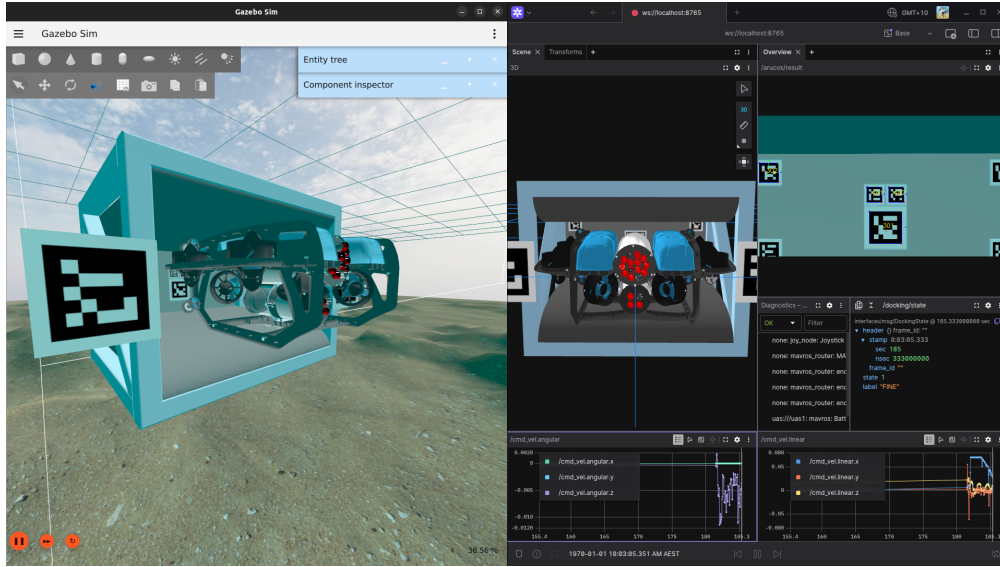


Figure 11: The simulation environment during a docking trial: Gazebo world view (left) and live telemetry in Foxglove (right), with the onboard camera stream, ArUco detections and the docking state machine state.

Dock swaying motion is generated by a custom Gazebo plugin that drives the dock through a sinusoidal sway and heave of amplitude of 0.1 m with 90-degree relative phase, producing an orbital motion representative of wave-driven forces on the moored structure. Mechanical capture and contact are not modelled such that success is defined at the capture envelope (Section 3.2.5) and potential dock contacts are post-processed during data analysis.

The swept period band and amplitude are mapped from the real wave forcing in New South Wales. The wave energy occupies roughly 5 to 20 s, and the mean sea state off the NSW coast is a wave height of 1.6 m at an 8 s period, in context of the operation of this work [11, 30, 16]. The dock is forced by the wave orbital motion, whose amplitude decays with depth exponentially: e^{-kz} with $k = 2\pi/L$ and deep-water wavelength $L = gT^2/2\pi \approx 1.56 T^2$ [11]. As a result, the simulated 0.1 m

amplitude corresponds, in the mean NSW sea state, to dock at approximately 19 m depth for the 6 s period, 33 m at 8 s, 52 m at 10 s and more than 200 m at 20 s. Therefore, sweeping 6 to 20 s at a fixed 0.1 m amplitude isolates the period dependence while keeping every trial identifiable with a physically realistic sea state in NSW and operating depth.

3.2.2 Marker layout

Each marker on the dock are sized and categorised by their function. The two largest markers (200 m class) sit on the outer front edge of the structure, where they subtend the pixels needed for long-range acquisition. The mid-sized and small markers populate the dock interior. The two smallest (IDs 401 and 402) are placed at the height of the vehicle’s camera in the docked position, so that markers remain in frame during the final phase of docking. This follows the fractal marker principle, in which markers of several sizes are combined so that pose estimation survives across the full range span [28], though the markers are distributed spatially instead of nested. The dock structure itself was kept deliberately minimal, and its mechanical design is not a subject of this thesis. The marker IDS are likewise arbitrary with respect to detection: they were assigned so that the leading digit encodes the size class (2xx outer, 3xx interior, 4xx eye level), which is a naming convention to ease debugging, and each ID merely selects a stock pattern from the dictionary with the least black and white pixel ratio. Optimising the bit patterns themselves for detectability at longer range is not considered in this thesis.

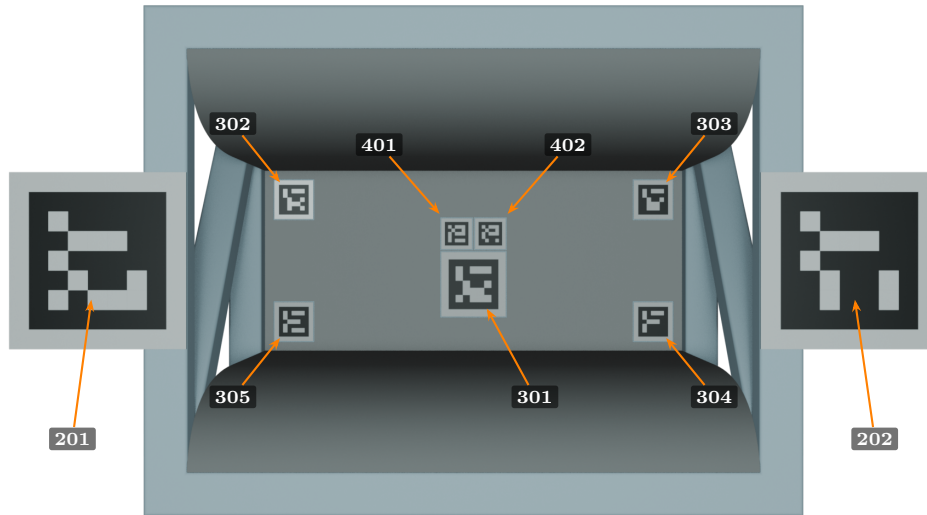


Figure 12: Dock front view with the size-graded marker placement: the 200 mm class (201, 202) on the outer edge, mid and small markers (301 to 305) in the interior, the smallest pair (401, 402) at camera eye level in the docked position.

3.2.3 Perception (simulated ArUco pipeline)

The full perception pipeline runs in the loop such that the controller never sees ground truth. Markers are detected on the simulated camera stream and each detection is converted to an implied dock-origin pose using the known marker layout. A spatial-consensus test rejects outlier detections, and the

surviving markers are fused by inverse-variance weighting with a range-scaled noise model, $\sigma = \alpha r^2/s$ for a marker of size s at range r , so distant small markers contribute less than near large ones.

The fused measurement feeds a nine-state error-state Kalman filter over dock position, dock velocity and a small-angle rotation error carried about a reference quaternion [32]. Two process-noise regimes are defined. The *static* regime pins the velocity states (zero initial velocity uncertainty and zero velocity process noise), which is algebraically equivalent to a constant-position filter and serves as the baseline. The *sway* regime models constant-velocity motion with a white-noise-acceleration process of density σ_a (0.16 m/s² nominal, exposed as a parameter), enabling dock velocity estimation and the feedforward of Section 3.2.4. Updates are gated by a Mahalanobis test, the velocity state is bounded by a physical dock-speed limit (0.2 m/s), and the filter publishes its velocity state as a twist alongside the pose.

A health classifier maps filter state to WARMING, HEALTHY, DEGRADED or STALE from the age of the last accepted measurement and the position uncertainty: healthy requires a recent update and position standard deviation under 0.04 m, and the estimate is declared stale when either the measurement age exceeds 3 s or the uncertainty exceeds a 0.15 m ceiling, so controllers hold rather than act on an unconstrained estimate. Downstream controllers consume only this semantic signal.

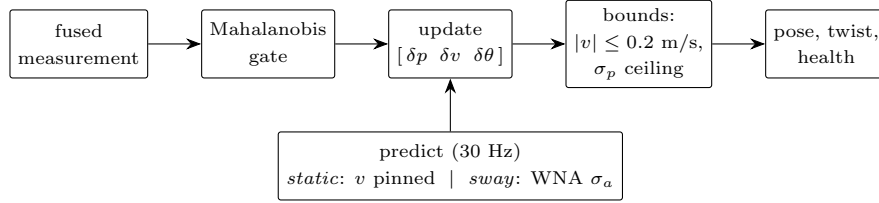


Figure 13: Filter structure: the nine-state error-state filter with the two process-noise regimes, the measurement gate and the state bounds.

3.2.4 Guidance and control

Coarse approach regulates a standoff point 1 m out along the dock entry axis with per-axis PD control, a distance-gated surge cap that bounds the worst-case closing speed, and the autopilot in depth-hold. Handoff to fine alignment is gated by position, boresight-offset and yaw tolerances plus a velocity-match term (an exponentially smoothed derivative of the standoff-relative position, threshold 0.15 m/s) sustained through a debounce, so the handoff cannot latch at a sway zero crossing where the position error is momentarily small while the dock moves fastest.

Fine alignment implements align-then-advance: surge toward the entry is inhibited until lateral, vertical and yaw errors are within alignment tolerances, with the autopilot in stabilise mode so the controller owns depth. The seat criterion (definition of docked) is a debounced 0.10 m range to the entry with 0.03 m lateral and vertical tolerances; the 0.10 m value is derived from the measured sensing envelope in Section 4.1, which shows marker visibility collapsing below approximately 5 cm, so the seat is confirmed where vision is still reliable and the final centimetres are delegated to the capture mechanism assumed by the envelope definition.

Both controllers accept a dock-velocity feedforward built from the filter velocity state rotated into the body frame. The feedforward is clamped dock-relatively (the feedback term retains its full authority

budget and the feedforward is limited separately at 0.25 m/s), scaled by a distance-gated authority ramp (zero beyond 1.5 m from the standoff, full inside 0.3 m), and converted from physical velocity to command units by a gain of 0.6 identified from the plant response: the command channel is effort-like, with a measured steady gain of 1.3 to 1.6 and approximately 240 ms of lag between command and vehicle velocity. In the static filter regime the velocity estimate is identically zero, so the feedforward vanishes and the system reduces exactly to the baseline controller; this is the ablation used throughout Part A.

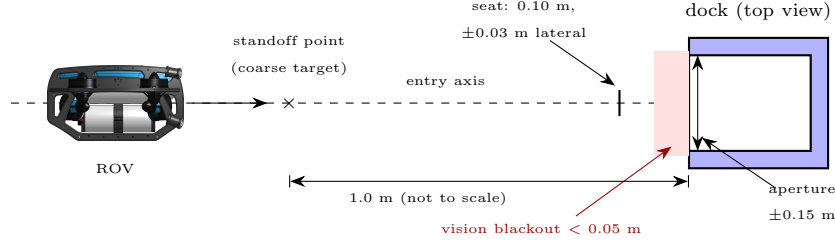


Figure 14: Terminal docking geometry (top view, not to scale): the coarse standoff point, the entry axis, the 0.15 m scoring aperture, the 0.10 m seat criterion and the close-range vision blackout.

3.2.5 Experimental method

Trials are executed by an unattended harness that launches the full stack, waits for the filter to initialise and the autopilot mode to settle, engages the state machine, records a curated set of topics, polls for the docked state with a timeout, and tears the stack down between trials. Recording runs in a low-impact configuration after measurements showed recorder load itself degrading the perception cadence it was capturing.

The envelope experiment sweeps filter regime (*static* baseline versus *sway* method) by dock sway period $\{20, 16, 12, 10, 8, 6\}$ s with three sway start phases per cell ($n=3$) plus a static-dock sanity cell per regime, 38 trials per sweep. Scoring uses the ground-truth channel only. Each trial receives one of three outcomes: CLEAN DOCK (docked with no pre-latch off-aperture crossing), CONTACT (the vehicle volume crossed the dock structure envelope more than 0.15 m from the calibrated entry axis before the seat latched; the trial is censored at that instant because subsequent motion traverses space a physical dock would deny), or NO DOCK. Crossing the entry plane inside the aperture is normal insertion and is not a contact. The aperture radius is a stated assumption standing in for a capture funnel; the two observed populations (centred insertions at 0.003 to 0.05 m offset versus misses at 0.17 m and beyond) are far from the boundary, so conclusions are insensitive to its exact value. The full sweep was run twice, on two hosts (real-time factors 0.6 and 1.0) and two filter revisions, to check that conclusions do not depend on load, hardware or incidental tuning.

3.3 Part B: Real-CV validation

3.3.1 Marker target and its as-printed geometry

The target uses the original ArUco dictionary (5×5 payload) with the dock’s marker set: two 155.64 mm markers (IDs 201, 202), one 77.82 mm (301), four 46.69 mm (302 to 305) and two 36.96 mm (401, 402),

printed across three A4 sheets taped to a rigid panel. The sheets were printed at 95.9% of nominal scale: a ruler measurement of marker 201 first revealed the shortfall, and a size-weighted least-squares fit over all nine markers recovers the same scale with residuals under 0.4 mm. The analysis therefore uses the measured black-square sides (149.4, 74.7, 44.4 and 35.6 mm) throughout; the nominal sizes would overstate every derived range by 4.1%. The as-measured layout is shown in Figure 15.

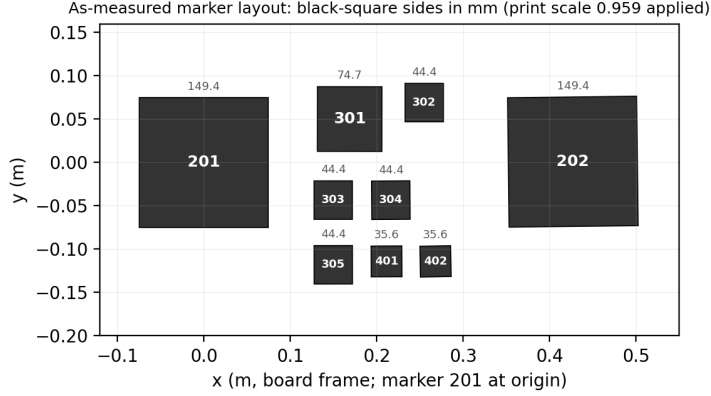


Figure 15: As-measured marker layout in the board frame, with positions and black-square sides estimated from the data at the 0.959 print scale. All four size groups appear in every frame at a common range.

3.3.2 Capture setup

Data was captured in a single pool session with a ZED stereo camera mounted on the BlueROV2, of which only the left, rectified stream was recorded (960×540 at nominal 15 Hz), giving a monocular ArUco problem; images, camera info and the ZED IMU were logged. The ZED was used in place of the BlueROV2’s built-in camera because its calibration had already been characterised for underwater use in prior work within the project group, and its driver publishes rectified images with the calibration alongside; the built-in camera had never been calibrated. The calibration was still verified against the captured data itself, as described below.



Figure 16: Capture platform: the BlueROV2 with the ZED stereo camera mounted for the pool session. Only the left lens was recorded.

Camera intrinsics are the shipped in-air factory rectification ($f_x=f_y=797.5$, zero distortion on the rectified stream). The in-water validity of this choice was tested by a joint bundle adjustment over 1400 marker observations in 317 frames, solving for underwater focal lengths, the coplanar board layout and per-frame poses with scale pinned by the ruler-measured marker sizes: the aspect ratio solves to $f_y/f_x = 0.996$ and the isotropic optimum is only 1.9% better than the in-air value, while the flat-port refraction correction of $1.333\times$ fits 44% worse and is rejected. The shipped intrinsics are therefore used unchanged, and all metric ranges carry an approximately $\pm 10\%$ scale uncertainty from the focal length, which dominates the error budget. For this reason the primary independent variable throughout Part B is the calibration-free *apparent marker size in pixels*; metric range is derived and reported with its uncertainty.

3.3.3 Experiment design

The two walk-backs sweep range continuously from about 0.6 m to beyond the detection limit. Marker size needs no separate sweep: the four size groups (149.4 down to 35.6 mm) are on the board simultaneously, so every frame samples all sizes at a common range and water condition. Viewing angle was not controlled: the on-axis run has a median marker incidence of 13° and the oblique run 58° , two disconnected regimes confounded with range and marker size, so no detection-versus-angle curve is reported.

No external ground truth exists (no tape or survey range, and vehicle odometry was not recorded); scale is anchored by the ruler-measured marker sizes and attitude by an IMU gravity check. Detection is scored per marker-frame trial against apparent-size bins with a minimum of ten trials per bin and Wilson confidence intervals (2551 trials on-axis, 950 oblique). Maximum detection range is established uncensored: the on-axis walk continued for a further 79 s (about 3 m) beyond the last detection with zero detections, and two independent end-of-run estimates (walk-back rate extrapolation and board

apparent width in the final frame) agree at 7.8 to 8.1 m, so the range limits reported below are true detection limits.

3.3.4 Turbidity measurement and synthetic degradation

Water attenuation was measured photometrically from the target itself, without dosing or a turbidity meter. A black patch (marker border) and a white patch (surrounding sheet) at the same range share an identical veiling-light term in the image formation model $I = J e^{-\beta d} + B(1 - e^{-\beta d})$, so their difference is free of B : $\text{contrast}(d) = (J_w - J_b) e^{-\beta d}$, and β is a straight-line fit in log space. The fit requires two corrections: an instrument-response correction $k(\text{apparent px})$, measured at fixed range so it is independent of β , without which small markers read up to 18% low and the bias mimics attenuation; and per-marker intercepts, since the three sheets sit under different local lighting. The underlying image formation model is the standard attenuation-plus-veiling form [13].

Higher turbidity than the pool’s was synthesised by re-attenuating each real frame in place: $I' = B + (I - B) e^{-\Delta\beta d(x)}$ per channel and per pixel, with the per-pixel range $d(x)$ taken from the board-plane pose and $\Delta\beta = m\beta$ for multipliers $m \in \{0, 0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 8\}$, where $m=0$ reproduces the real frame. Results are reported against total optical depth $\tau = (1+m)\beta d$ rather than named water types: range enters the model only as the product βd , so any uniform scale error in d is absorbed into β ; the τ axis is therefore unaffected by the $\pm 10\%$ range uncertainty and comparable across waters and cameras. The synthesis adds no scattering blur, so any benefit a contrast-enhancing preprocessor shows on synthesised frames is an upper bound. The full chain is summarised in Figure 17.

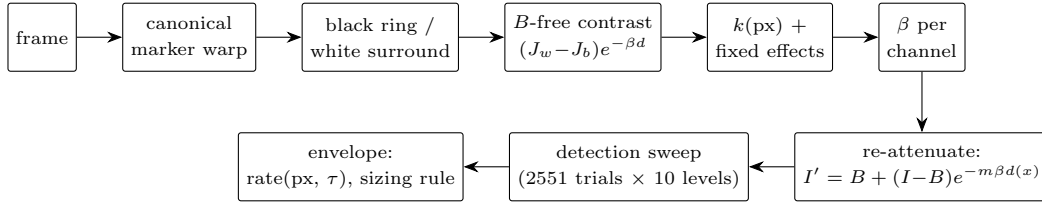


Figure 17: Turbidity processing: the top row measures the water’s attenuation from the target itself; the bottom row stresses the same frames to higher optical depth.

3.3.5 Detection pipeline and metrics

Detection uses OpenCV 4.10 with the current `ArucoDetector` API, subpixel corner refinement, and `adaptiveThreshConstant` lowered from the default 7 to 3, the best of a grid over $\{3, 5, 7\}$ scored at the highest synthesised turbidity (selected without a holdout, so its reported rate is an optimistic bound). Single-marker pose, the quantity under test, uses planar PnP (IPPE square); the reference for self-consistency is a multi-marker board PnP computed leave-one-out, with the marker under test excluded from its own reference. Off-board IDs are not filtered, so the misidentification rate remains measurable. Reported metrics are detection rate per apparent-size bin and per derived-range bin, uncensored maximum detection range per marker, single-marker translation and rotation deviation from the board reference (self-consistency, not accuracy, since no ground truth exists), misidentification rate, and an IMU gravity check as the sole camera-independent attitude validation.

4 Results and discussion

4.1 Part A: Simulation results

4.1.1 Perception envelope

The filter’s accuracy is governed by range through the $\alpha r^2/s$ measurement noise model. With the vehicle stationary 5 m from the dock the fused noise floor is approximately 0.11 m and the filter reports DEGRADED throughout; at 2.2 m the floor drops to approximately 0.02 m, the filter is HEALTHY for 92% of the run (median position standard deviation 0.018 m), and the velocity state tracks the true dock velocity with amplitude ratio 0.85 and correlation 0.89 (Figure 18). At the opposite extreme, marker visibility collapses at contact range: binning fine-alignment health by ground-truth range shows 84% to 100% healthy beyond 5 cm of the entry but only 21% inside it, a blackout produced by marker corners overflowing the camera field of view. This is the size-graded layout of Section 3.2.2 working as designed: detection hands over from the outer large markers to the interior and eye-level markers as each size class successively overflows, and the blackout begins only where even the smallest pair no longer fits the frame. The measured envelope, reliable from roughly 2.5 m down to 0.05 m, motivates both the 0.10 m seat criterion and the delegation of the final centimetres to mechanical capture.

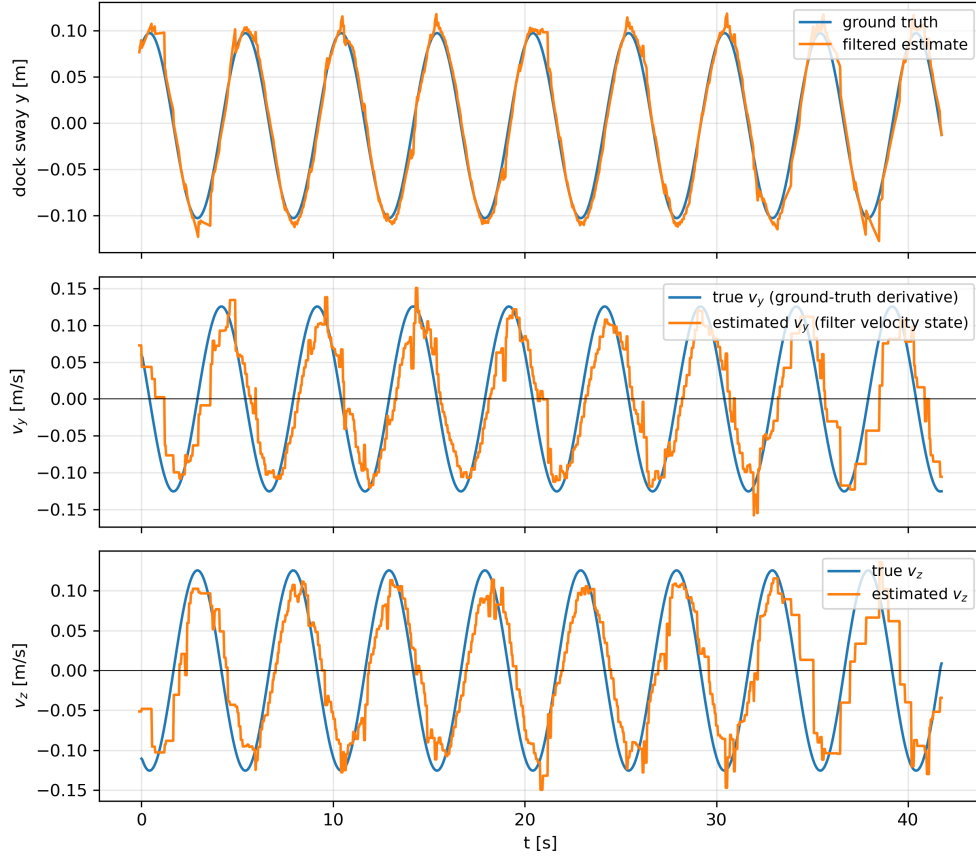


Figure 18: Dock velocity estimation with the vehicle stationary at 2.2 m (5 s sway period). Top: filtered dock sway against ground truth. Middle and bottom: the filter velocity state against the ground-truth dock velocity in sway and heave.

4.1.2 Docking demonstration

Against the nominal 18 s, 0.1 m orbital sway, the system docks autonomously end to end (Figure 19): engagement to coarse at $t+4$ s, handoff to fine alignment at $t+15$ s, seated and docked at $t+26$ s with no demotions, latching 7 to 8 cm from the entry and coming to rest 4 cm in front of the dock plane. The trajectory shows the two-stage structure directly: the coarse phase closes 4 m to the standoff point, and fine alignment performs the lateral correction onto the moving entry axis before the final advance. The range curves plateau briefly at 1 m in both trials: this is the hold at the coarse standoff, where the vehicle decelerates under the distance-gated surge cap, overshoots the standoff point slightly and settles back while the debounced handoff gate clears; in the sway trial the dock’s own motion during the hold adds a small range wobble on top. The static-dock sanity trial is overlaid for comparison: it follows an essentially straight approach and seats at 19 s, so the cost of the 18 s sway is the lateral correction and roughly seven additional seconds, not a qualitative change in behaviour. Both trials, and every sweep trial below, ran the full perception pipeline in the loop with no ground-truth input to the vehicle.

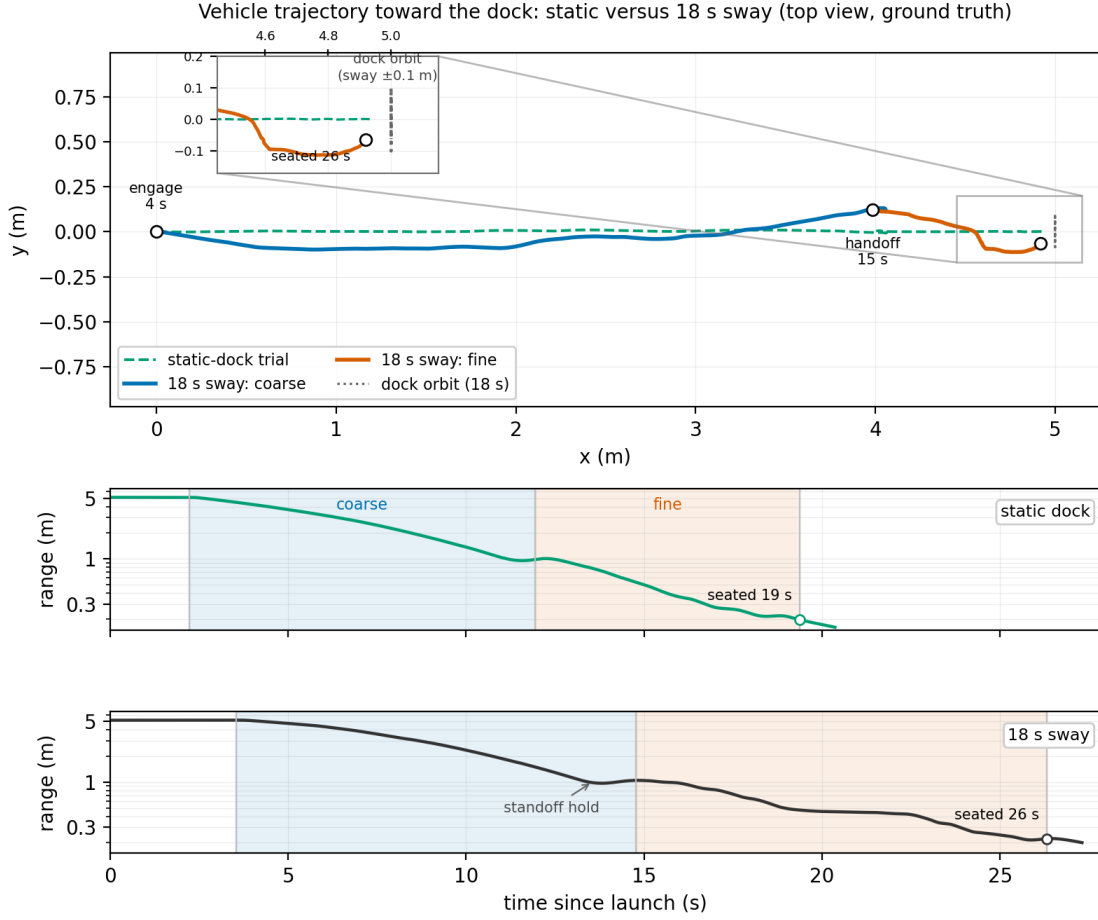


Figure 19: The demonstration trial (18 s sway period) against the static-dock sanity trial. Top: ground-truth vehicle trajectories in the horizontal plane, coloured by mission phase, with the terminal region magnified in the inset. Bottom: range to the dock origin on a logarithmic scale, one panel per trial with its phase timing shaded. The plateau at 1 m is the hold at the coarse standoff.

4.1.3 Docking envelope versus sway period

Figure 20 shows the result of the sweep ($n=3$ per cell, contact-censored scoring). The baseline (static filter, no feedforward) docks in every trial at every period in the raw scoring, and cleanly down to 8 s; at 6 s, two of three trials brush the aperture boundary at low speed (0.03 to 0.06 m/s) before latching. The method (constant-velocity filter with feedforward) matches the baseline down to 12 s, degrades at 10 s (one pre-latch contact), and fails at 8 s and 6 s; its single raw success at 8 s was contact-assisted.

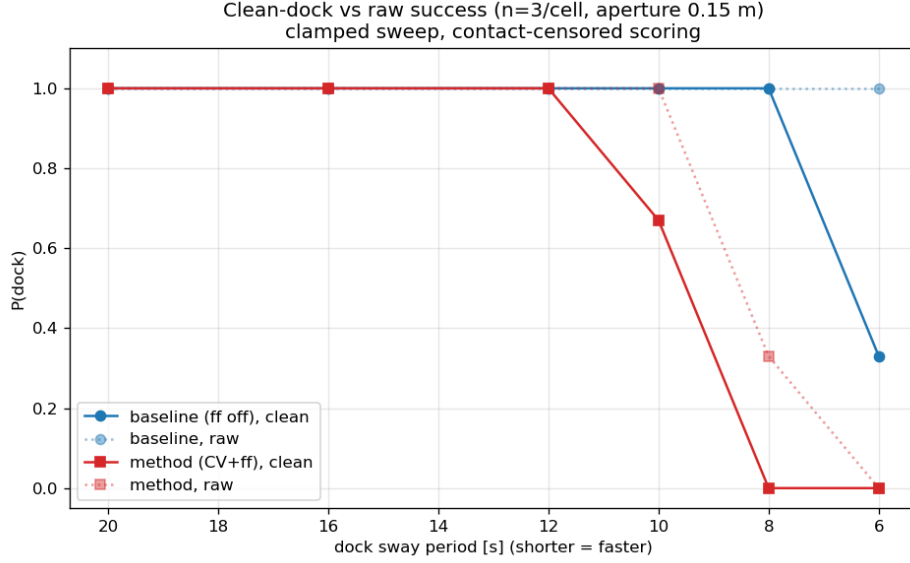


Figure 20: Docking success versus sway period under raw (dotted) and contact-censored clean (solid) scoring.

In the operational terms of Section 3.2.1, each period cell maps to a sea state and dock depth. The mean NSW sea (8 s period, 1.6 m significant height) forces a 0.1 m sway on a dock near 33 m depth, which the baseline handles cleanly; the 6 s wind-sea cell corresponds to a shallow dock near 19 m in the same sea, where the baseline still docks in every trial under raw scoring; and the longer-period cells correspond to deeper moorings reached only by swell, where both configurations succeed. The measured envelope therefore covers moored docks across typical continental-shelf operating depths, with residual risk concentrated in shallow, short-period deployments, and the method’s cliff at 8 to 10 s sits at the region’s mean wave period, which is precisely why the baseline’s structural immunity matters operationally.

Clean insertions cross the entry plane 0.003 to 0.05 m off axis at 0.06 to 0.11 m/s closing speed, and all observed contacts occur below 0.06 m/s. These figures form the interface requirement for a capture mechanism: a funnel with a capture radius of at least 0.15 m and tolerance for approximately 0.1 m/s closing speed captures everything the baseline delivers down to 8 s periods.

4.1.4 Failure analysis: ego-motion instability of the velocity estimator

The method’s fast-period failures trace to a single mechanism. The dock measurement is reconstructed in the world frame through the transform of a moving camera, so a fraction of the vehicle’s own motion leaks into the measured dock position. This leakage is *correlated with the vehicle state*, violating the filter’s white-noise assumption. The static filter, with no velocity state, absorbs it as position noise and remains faithful (estimated sway amplitude 0.7 to 0.9 times truth in all baseline runs). The constant-velocity filter integrates it into a phantom dock velocity and extrapolates: estimated amplitude reaches 2.4 to 5 times truth in successful method runs and 18 to 52 times truth in the failures, where the estimate has decoupled from the real dock (correlation 0.00 with the true dock

while the estimate's excess motion correlates at 0.88 with the vehicle's own position). The feedforward closes the loop by injecting the phantom velocity directly into the command; with the identified plant lag of approximately 240 ms, the loop gain crosses unity between the 10 s and 8 s periods, which is where the measured cliff sits.

Figure 21 shows the mechanism in a single failed trial at the 8 s period. In the tracking segments the estimated dock sway already exceeds the true amplitude while the vehicle overshoots further still, the chase visible as the vehicle's lateral excursion enclosing the dock's. When the resulting misalignment costs marker visibility and measurements are gated out, the constant-velocity model dead-reckons on the phantom velocity, ramping the estimated dock position to a full metre off (ten times the true amplitude, with the velocity clamp active) while the vehicle follows; reacquisition snaps the estimate back, the supervisor demotes, and the cycle repeats without ever seating.

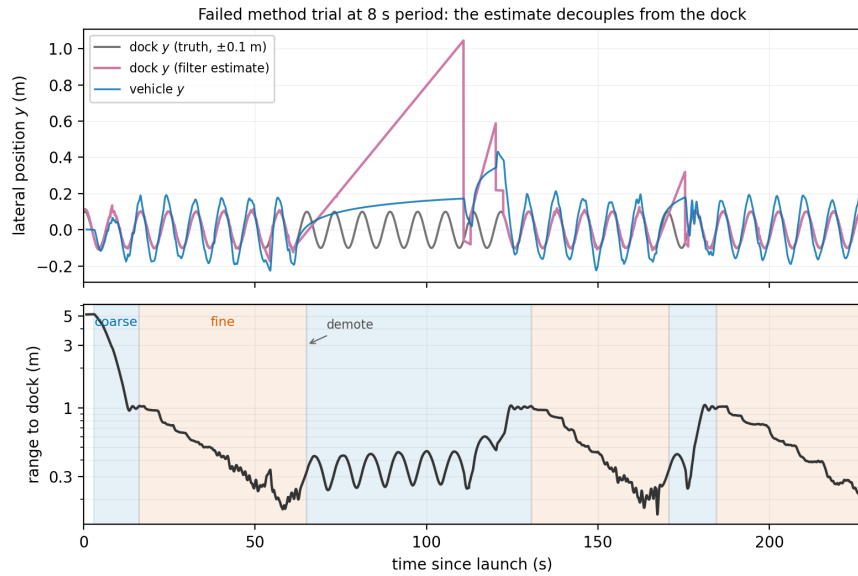


Figure 21: A failed method trial at the 8 s sway period. Top: lateral position of the true dock, the filter's dock estimate and the vehicle. Bottom: range to the dock with mission phases shaded.

The failure is self-concealing: every alignment signal the system exposes is computed against its own estimate, and in the failed trials those signals read seat-tolerance alignment (lateral errors of 0.000 to 0.017 m) while ground truth shows the vehicle 0.17 to 0.48 m off axis at the dock plane. Only the ground-truth channel exposes the divergence, an argument for independent verification in any real deployment.

Whether this failure would transfer to hardware turns on whether the 240 ms plant lag is real or a simulation artefact, since the loop-gain argument places the cliff where the lag becomes comparable to the sway timescale. It is not a compute artefact: the sweep replicated at real-time factors of 0.6 and 1.0 with the cliff in the same place, so the lag does not scale with simulation load. Nor is its origin simulator-specific. The autopilot in the loop is the same ArduSub firmware a physical BlueROV2 runs, executed software-in-the-loop, so the path from velocity command through mode controller to motor mixing is the production code path, and the remainder of the lag is the physics of accelerating the

vehicle’s inertia and added mass under bounded thrust, both taken from the vehicle’s hydrodynamic model rather than chosen for the simulator. If anything the simulation is optimistic: a real vehicle adds ESC and thruster response, the autopilot’s own sensor filtering, and tether drag, none of which are modelled, so the hardware lag should be equal or larger. The instability would therefore appear on the real vehicle at the same or longer sway periods, and a hardware feedforward design cannot assume the lag away.

Three progressively stronger mitigations were evaluated. Bounding the velocity state at the physical dock speed and adding the covariance ceiling eliminated the metre-scale runaway and the resulting collisions, but did not restore fast-period docking. Sweeping the process noise density σ_a over 0.02 to 0.16 m/s² found no stable setting: estimated amplitude remained 5 to 37 times truth at every value with no monotonic trend, because a smaller gain also makes the filter slower to shed an acquired phantom velocity. The baseline’s immunity is structural rather than a tuning point: its velocity gain is exactly zero, and no finite gain interpolates to that property while the measurement error remains correlated with vehicle motion. The conclusion is that dock velocity feedforward from a world-frame constant-velocity estimator requires explicit ego-motion compensation, by fusing vehicle odometry into the filter or estimating in a vehicle-relative frame, identified as future work.

4.1.5 Replication

The sweep was run in full on two hosts (real-time factors 0.6 and 1.0) and two filter revisions (before and after the velocity bounds). The baseline’s uniform success and the method’s cliff between 10 s and 8 s reproduced in both, indicating the result reflects the estimator design rather than compute load, hardware or incidental tuning.

4.2 Part B: Real-CV results

4.2.1 Detection rate versus marker size and range

Detection is governed by apparent size in pixels with a sharp floor. Across all four size groups the rate is zero below 21 px, transitional in the 21 to 28 px bin (0.21 to 0.59 depending on size), and above 0.85 from 38 px upward (Figure 22); the ~ 27 px 50% budget measured independently in Section 4.2.5 crosses each group’s curve near its 0.5 point. On the derived metric axis this maps to overall on-axis rates of 0.93 for the 149.4 mm markers, 0.61 at 74.7 mm, 0.43 at 44.4 mm and 0.29 at 35.6 mm over the swept range. The oblique run (median incidence 58°) detected only the 149.4 mm markers, at 0.60 overall; the three smaller groups produced zero detections in 791 trials, indicating that oblique viewing costs roughly one size class even before the angle reaches the detector’s geometric limits.

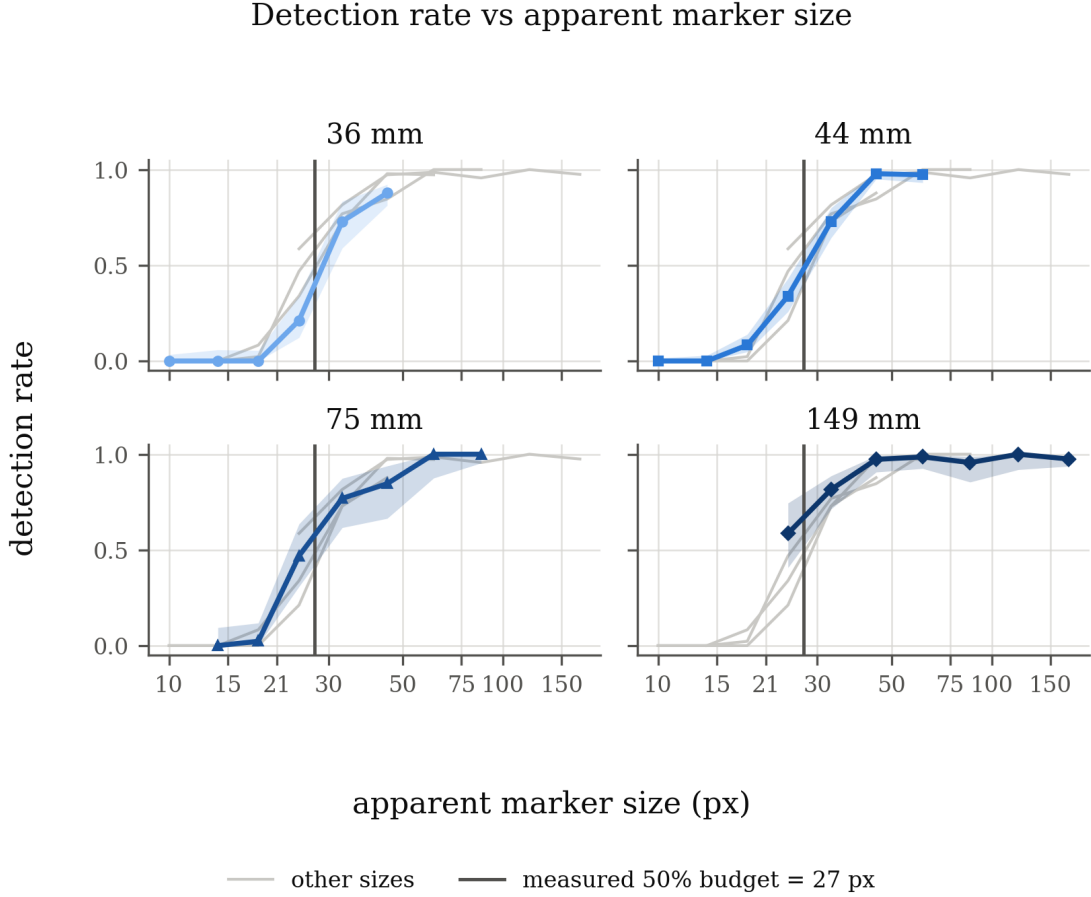


Figure 22: Detection rate versus apparent marker size per size group, with Wilson 95% intervals; bins with fewer than ten trials suppressed. The vertical line is the ~ 27 px 50% budget (Section 4.2.5).

4.2.2 Maximum detection range

Uncensored maximum detection ranges scale linearly with marker side: 5.10 and 4.87 m for the two 149.4 mm markers, 3.05 m at 74.7 mm, 1.55 to 2.17 m at 44.4 mm, and 1.10 to 1.14 m at 35.6 mm, fitted by $\text{max range} \approx 34.9 \times \text{side}$ (Figure 23). The walk-back continued approximately 3 m past the last detection with none recorded, so these are genuine detection limits. Two distinctions matter operationally. First, only the largest markers reach the 5 m acquisition range assumed for the coarse phase, and only marginally. Second, maximum range is a single-tail-detection statistic; the *reliable* (50%) range sits lower, at 5 m a 149.4 mm marker subtends 23.8 px, below the ~ 27 px needed for 50% detection (Section 4.2.5), which is why its maximum range of 5.1 m coexists with a 50% range nearer 4.3 m. Only the second number sizes a dock. The law also validates the size assignment of the dock layout (Section 3.2.2): the largest markers alone carry long-range acquisition, while the interior small markers need only the close-range band they were placed for.

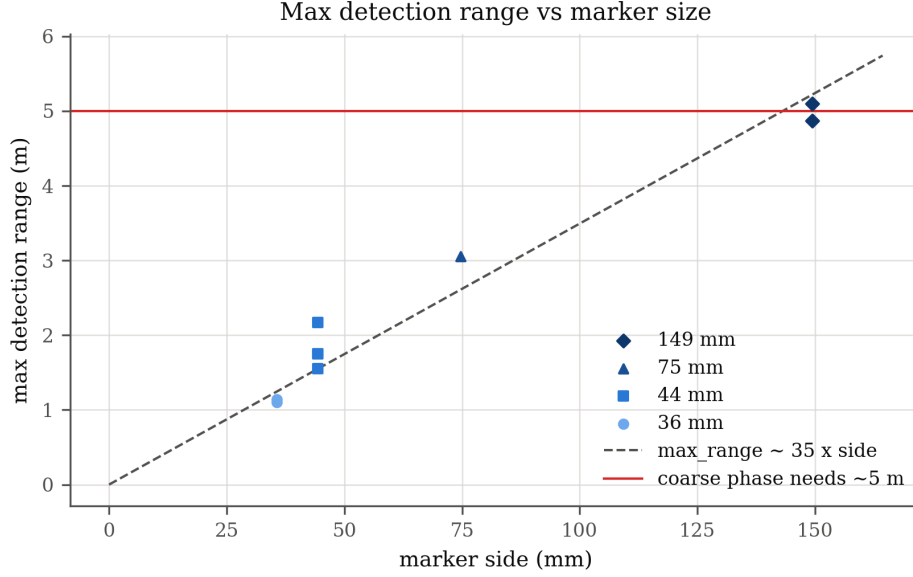


Figure 23: Maximum detection range versus marker side with the fitted linear law and the 5 m coarse-acquisition reference line.

4.2.3 Pose self-consistency and misidentification

With no external ground truth, single-marker pose is scored against a leave-one-out multi-marker board reference, a self-consistency measure, not accuracy. Translation deviation grows from 10.6 mm mean (9.4 mm jitter) in the 0.5 to 1.0 m bin to 36.1 mm mean (32.9 mm jitter) at 2.0 to 3.0 m; rotation deviation is 11 to 16° and roughly flat with range. These figures pool all marker sizes, since a per-size split would change the reference geometry with the marker under test and read backwards. The sole camera-independent validation is an IMU gravity check: over 323 frames the board’s estimated tilt from vertical has a median of 6.6°, consistent with a hand-held, approximately vertical board. A yaw-rate cross-check was not corroborative (only 7 of 25 turn segments had sufficient board coverage, and vision magnitudes did not track the gyro), and is reported as such. Misidentification was essentially absent in real water: one off-board ID in 1376 on-axis detections (rate 0.0015) and none in the oblique run.

4.2.4 Attenuation measurement

The B-free contrast fit yields, with instrument correction and per-marker intercepts over 934 samples spanning 0.62 to 3.19 m: $\beta_b = 0.279$, $\beta_g = 0.247$, $\beta_r = 0.443$, $\beta_{grey} = 0.289 \text{ m}^{-1}$ ($r^2 = 0.71, 0.67, 0.88, 0.75$; Figure 24). The strongest support for the correction chain is a quantity it never saw: the red-to-green attenuation ratio moves from 1.50 (raw) to 1.79 (corrected), against approximately 1.8 expected from water optics. A residual check over 5629 near-to-far marker pairs (not a statistical cross-validation, no holdout was used) shows the exponential form predicts real far-frame contrast to a median error of −3.1% corrected, versus +18.5% raw. At 5 m this water gives a grey optical depth $\tau = 1.44$, i.e. 23.6% of contrast surviving. The fitted veiling light B is explicitly not physically

meaningful (negative r^2 ; the green estimate exceeds the white sheet’s own brightness): the method is built so that B cancels from contrast, and B enters the synthesis only as a DC term that adaptive thresholding rejects.

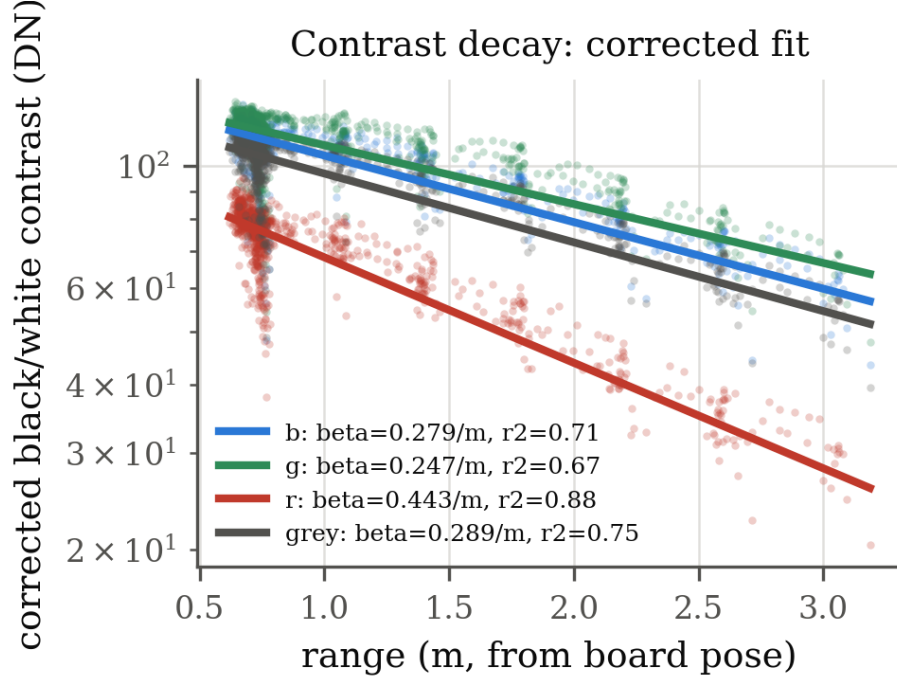


Figure 24: Instrument-corrected black-to-white contrast versus range with the fixed-effects exponential fits per channel.

4.2.5 Turbidity envelope and marker sizing

Re-attenuating all 2551 on-axis trials at ten turbidity multipliers (a fixed denominator at every level) maps the detection envelope over apparent size and total optical depth (Figure 25). The envelope has two regimes: below $\tau \approx 2$ the boundary is vertical, detection is *pixel-limited* and turbidity hardly moves it; above $\tau \approx 2.3$ the boundary is horizontal, detection collapses at every apparent size sampled, *turbidity-limited*. The pixel budget quantifies the first regime: the apparent size needed for 50% detection is 27.3 to 28.1 px, flat across a sevenfold span of optical depth (τ 0.18 to 1.30), rising only within the resolution of the binning before the collapse (Figure 26). The conclusion is that ArUco’s adaptive thresholding makes detection remarkably insensitive to contrast loss until an abrupt threshold near $\tau \approx 2$, a cliff, not a gradient. Apparent size and turbidity are the same pair of governing variables identified by the tank evaluation of dos Santos Cesar et al. [6], which reported the smallest detectable size growing with turbidity at six discrete levels; the τ -resolved envelope here refines that picture into a flat pixel-limited regime followed by an abrupt collapse.

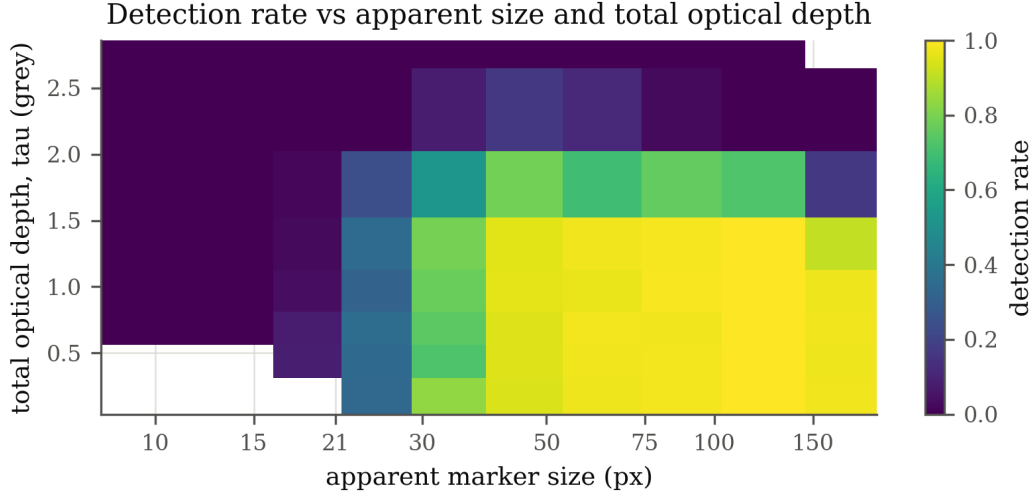


Figure 25: Detection rate over apparent size and total optical depth; white cells are unreachable combinations.

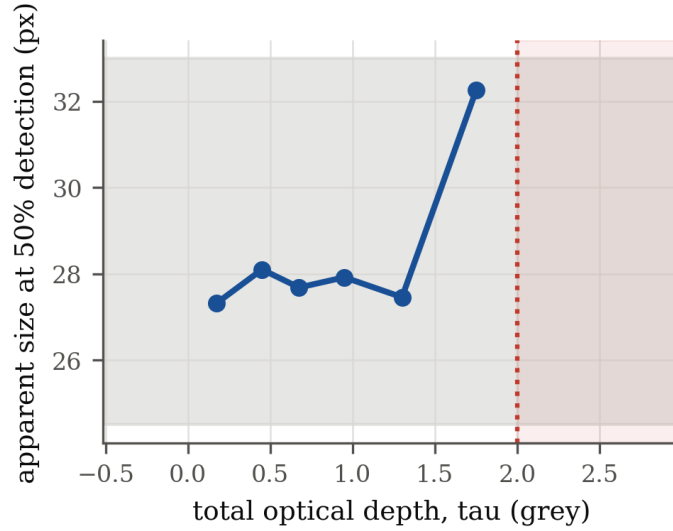


Figure 26: Apparent size required for 50% detection versus total optical depth: flat at ~ 27 px up to $\tau = 1.3$. Above $\tau = 2$ (dotted line, shaded region) no measured size reaches 50%. The grey band is the single pixel-bin interval (24.5 to 33.0 px) within which the crossing is interpolated, so no trend narrower than the band is resolvable.

Inverting the pixel budget through the pinhole model sizes the dock marker: for 50% single-frame detection this pool's water requires a 105 mm marker at 3 m and 182 mm (interval 154 to 207 mm) at 5 m. In clearer water ($\beta = 0.15 \text{ m}^{-1}$) the 5 m requirement barely drops (174 mm), confirming that below the cliff the requirement is set by pixels, not water; in murkier water ($\beta \geq 0.6 \text{ m}^{-1}$, $\tau \geq 1.8$ at 3 m) the requirement is undefined within the measured envelope, no marker size sampled achieves 50%. The synthesised degradation is illustrated in Figure 27; a forward-scatter check found no measurable

edge blurring within apparent-size bins across the synthesised range (slopes scatter around zero), so at these optical depths contrast loss, not blur, is the operative mechanism.

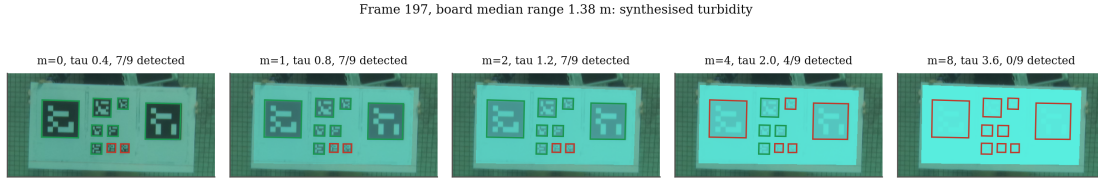


Figure 27: One frame re-attenuated at increasing turbidity multipliers, detections outlined: 7 of 9 markers at the real condition, 0 of 9 at the highest multiplier. The highest multipliers are diagnostic only; their DC term is unphysical.

4.2.6 Preprocessing: CLAHE evaluated and rejected

CLAHE was evaluated against the detector’s own thresholding knob over all 2551 trials at every turbidity level. In *real* water (multiplier 0) CLAHE is a net negative: detection 0.512 versus 0.519 baseline, while introducing 0.022 misidentifications per frame where the baseline has zero. Its apparent gain at high synthesised turbidity (0.337 versus 0.159 at the highest level) is measured on frames whose degradation adds no scattering noise, and is therefore an upper bound unproven for real water. Lowering `adaptiveThreshConstant` to 3 recovers about two-thirds of that gap (0.276) at a third of CLAHE’s misidentification rate (Figure 28). The engineering decision follows the real-water evidence: no CLAHE stage in the pipeline; tune the detector’s threshold constant instead.

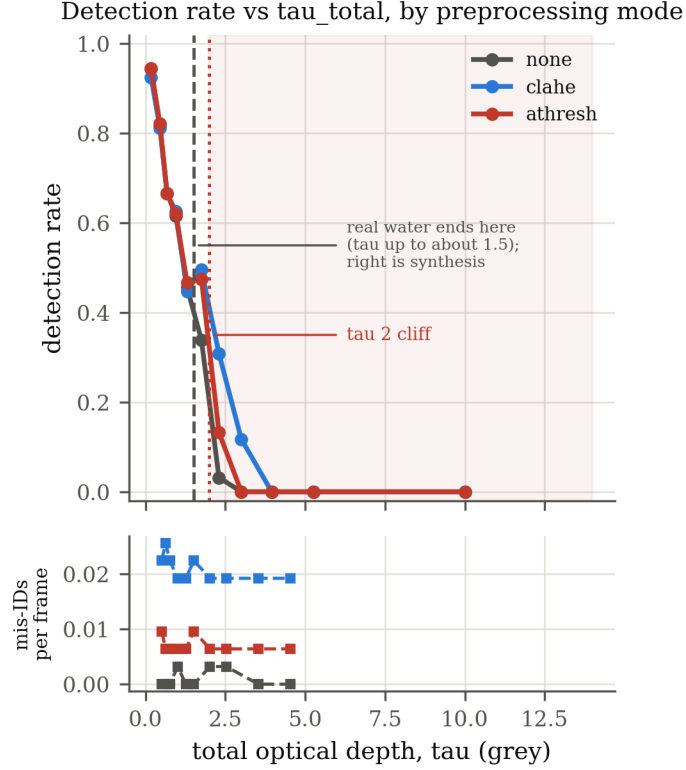


Figure 28: Detection rate (top) and misidentifications per frame (bottom) versus total optical depth for no preprocessing, CLAHE, and a lowered adaptive-threshold constant. Real water ends at the dashed line; everything beyond is synthesised.

4.2.7 Limitations

The dataset is one pool, one session: the optical-depth axis is designed to generalise (results are reported against $\tau = \beta d$), but this water is a single point and the attenuation spectrum of other waters will differ. No external ground truth exists; pose results are self-consistency only, and all metric ranges carry $\pm 10\%$ from the in-air focal length. β was fitted over 0.62 to 3.19 m, so quoting τ at 5 m assumes β constant with range. The oblique run is thin (161 detections, largest markers only) and viewing angle was never controlled. The capture was hand-held, so motion blur is present and unmodelled, and synthesised turbidity levels derive from the same frames and are correlated samples, so the quoted confidence intervals understate uncertainty at high multipliers.

4.3 Synthesis

The two halves bound the operating envelope of passive-marker terminal docking from opposite sides. Part A bounds the *dynamics*: with the full perception pipeline in the loop, the vehicle captures a swaying dock cleanly down to 8 s periods at 0.1 m amplitude, limited by its own control bandwidth (240 ms plant lag), and delivers insertions a capture funnel of 0.15 m radius and 0.1 m/s tolerance can accept. Part B bounds the *sensing*: a marker is reliably detectable only above ~ 27 px of apparent

size and below $\tau \approx 2$ of optical depth, which in the measured water prices the 5 m acquisition range at a 182 mm marker (154 to 207 mm) and rules out reliable acquisition at that range in substantially murkier water regardless of size.

The intersection is the design statement for the dock. The 200 mm-class markers assumed throughout Part A are validated by Part B as marginally sufficient for 5 m acquisition in pool-clarity water, with the per-size fusion pipeline and size-graded layout of Part A (Section 3.2.2) consuming exactly the size-dependent envelope Part B measured (small markers taking over as large ones overflow the field of view at close range, whose 5 cm blackout sets the seat criterion). Beyond the measured envelope, at longer standoff or higher turbidity, the passive-marker channel provides no signal, and terminal docking must be handed a vehicle already inside the envelope by a far-field cue such as an active light beacon, which is precisely the acquisition stage this work scoped out and now quantifies the need for.

5 Conclusion

This thesis built and validated the terminal stage of underwater docking from passive markers and classical computer vision, the approach open to low-cost platforms: multi-marker ArUco fusion, an error-state Kalman filter with two dock-motion models, and a coarse-to-fine docking state machine. The full perception pipeline runs in the loop, and the vehicle never receives ground truth.

In simulation the system docks onto a moving target. The baseline configuration docked at every sway period down to 8 s at 0.1 m amplitude, which covers moored docks at typical continental-shelf depths in the mean New South Wales sea state. The velocity feedforward fails at short periods because the vehicle’s own motion leaks into the dock measurement and is integrated into a phantom dock velocity. Tuning cannot fix it, the system’s own health signals cannot see it, and the baseline is immune because it never estimates velocity.

On real underwater imagery, detection needs about 27 pixels of apparent marker size, maximum detection range is about 35 times the marker side, and detection collapses once the total optical depth exceeds about 2. In the measured pool water this sets the marker for 5 m acquisition at 182 mm (154 to 207 mm).

Together the two parts bound where this class of system works, from the motion side and from the sensing side, and they supply the numbers a dock designer needs: a capture funnel of at least 0.15 m radius tolerating about 0.1 m/s closing speed, and the marker sizing rule above. Both are measured quantities. No stage of the pipeline depends on a trained model, so the measured limits are properties of the approach itself.

5.1 Future work

The immediate priority is the velocity estimator. Fusing vehicle odometry into the filter, or estimating in a vehicle-relative frame, would remove the ego-motion coupling at its source; any redesign must respect the 240 ms plant lag that sets where the instability appears. The natural companion step is hardware: Part B measured the camera side open loop, and the next experiment is to close the loop on the real BlueROV2 in a pool.

On the dock side, the capture funnel can now be designed against the delivered interface numbers, and contact needs a real mechanical model, since the simulation’s collision model is only demonstration-grade. The search phase would benefit from an active light cue: markers barely reach 5 m at practical sizes, and the linear range law makes larger markers an expensive way to buy range.

On the measurement side, an underwater camera calibration would remove the $\pm 10\%$ range uncertainty that dominates Part B’s error budget, and the detection measurements should be repeated in genuinely turbid water with a holdout, since the synthetic-turbidity results are an upper bound until validated there. Finally, a sway-aware dock motion model, narrowband or harmonic in place of constant velocity, may recover the feedforward benefit without the ego-motion sensitivity.

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Appendix: dock pose measurement and filter formulation

This appendix records the perception components that were modified from their published forms, as implemented in the released code. The standard error-state machinery (quaternion error composition, inject and reset) follows Solà [32] and is not rederived. Quaternion products are written $\otimes$, $R(q)$ is the rotation matrix of quaternion q , and $\text{Log}(\cdot)$ maps a quaternion to its rotation vector.

A.1 Per-marker dock pose and measurement noise

Each detected marker i yields a PnP pose $(\mathbf{p}_i^c, q_i^c)$ in the camera frame. With the marker’s surveyed pose in the dock frame $(\mathbf{p}_i^d, q_i^d)$ known from the layout, the dock origin implied by that single marker is

$$q_{\text{dock},i}^c = q_i^c \otimes (q_i^d)^{-1}, \quad \mathbf{p}_{\text{dock},i}^c = \mathbf{p}_i^c - R(q_{\text{dock},i}^c) \mathbf{p}_i^d. \quad (1)$$

The measurement noise assigned to marker i follows the PnP error scaling: position error grows as $r^2/(sf_{px})$ for range r , marker side s and focal length f_{px} , while rotation error, set by angular rather than metric resolution, carries one factor of r less. With the fixed camera intrinsics absorbed into a single calibrated constant α ,

$$\sigma_{p,i} = \alpha r_i^2/s_i, \quad \sigma_{R,i} = \alpha r_i/s_i, \quad \mathbf{R}_i = \text{diag}(\sigma_{p,i}^2 \mathbf{I}_3, \sigma_{R,i}^2 \mathbf{I}_3), \quad (2)$$

with $\alpha = 0.001$ in all Part A trials. This range-squared kernel replaces the exponential weight of Xu et al. [37], which saturates numerically once r/s leaves its calibrated band; the quadratic form grows without bound at far range, which is physically correct and handled downstream by the validation gate.

A.2 Spatial consensus

Before fusion, the implied dock orientations are checked for mutual consistency following the spatial outlier test of Kim et al. [15]. For each pair (i, j) the geodesic rotation distance $\theta_{ij} = \left\| \text{Log} \left((q_{\text{dock},i}^c)^{-1} \otimes q_{\text{dock},j}^c \right) \right\|$ is compared against a threshold of 8° ; each marker’s disagreement count is the number of pairings exceeding the threshold, and the worst marker is removed iteratively until all survivors agree. The test is skipped when fewer than three markers are visible, since no majority exists.

A.3 Inverse-variance fusion

The surviving single-marker poses are fused by inverse-variance weighting [37]. Position fuses in closed form,

$$\mathbf{P}_p = \left(\sum_i \sigma_{p,i}^{-2} \mathbf{I}_3 \right)^{-1}, \quad \hat{\mathbf{p}} = \mathbf{P}_p \sum_i \sigma_{p,i}^{-2} \mathbf{p}_{\text{dock},i}^c, \quad (3)$$

and orientation fuses by the same weights in the tangent space at the first survivor’s quaternion q_1 : with $\boldsymbol{\delta}_i = \text{Log}(q_1^{-1} \otimes q_{\text{dock},i}^c)$,

$$\mathbf{P}_R = \left(\sum_i \sigma_{R,i}^{-2} \mathbf{I}_3 \right)^{-1}, \quad \hat{q} = q_1 \otimes \text{Exp} \left(\mathbf{P}_R \sum_i \sigma_{R,i}^{-2} \boldsymbol{\delta}_i \right), \quad (4)$$

valid to first order because the consensus test bounds the tangent deltas. The fused pose, its covariance and the surviving marker count form the measurement passed to the filter.

A.4 Error-state Kalman filter

The filter state is $\mathbf{x} = [\mathbf{p}, \mathbf{v}, \delta\boldsymbol{\theta}] \in \mathbb{R}^9$: dock position, dock velocity and a small-angle rotation error about a reference quaternion q_{ref} carried alongside the state [32]. The prediction at rate 30 Hz is constant-velocity in position and identity in rotation,

$$\mathbf{p} \leftarrow \mathbf{p} + \mathbf{v} \Delta t, \quad \mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 \end{bmatrix}, \quad \mathbf{P} \leftarrow \mathbf{F} \mathbf{P} \mathbf{F}^T + \mathbf{Q}. \quad (5)$$

The two process-noise regimes of the Part A ablation differ only in $\mathbf{Q}$. The *static* regime is a position random walk with the velocity block zeroed,

$$\mathbf{Q}_{\text{static}} = \text{diag}(q_p \Delta t \mathbf{I}_3, \mathbf{0}_3, q_R \Delta t \mathbf{I}_3), \quad q_p = q_R = 10^{-4}, \quad (6)$$

while the *sway* regime is white-noise acceleration of density σ_a (nominally 0.16 m/s²) with the standard coupled blocks per axis,

$$\mathbf{Q}_{\text{sway}} = \begin{bmatrix} \sigma_a^2 \frac{\Delta t^3}{3} \mathbf{I}_3 & \sigma_a^2 \frac{\Delta t^2}{2} \mathbf{I}_3 & \mathbf{0} \\ \sigma_a^2 \frac{\Delta t^2}{2} \mathbf{I}_3 & \sigma_a^2 \Delta t \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & q_R \Delta t \mathbf{I}_3 \end{bmatrix}. \quad (7)$$

The fused measurement observes position and orientation directly, $\mathbf{H} = [\mathbf{I}_3 \mathbf{0} \mathbf{0}; \mathbf{0} \mathbf{0} \mathbf{I}_3]$, with innovation $\mathbf{y} = [\hat{\mathbf{p}} - \mathbf{p}, \text{Log}(q_{\text{ref}}^{-1} \otimes \hat{q})]$ and the fused blocks as $\mathbf{R}$. An update is accepted only if it passes the validation gate

$$d^2 = \mathbf{y}^T (\mathbf{H} \mathbf{P} \mathbf{H}^T + \mathbf{R})^{-1} \mathbf{y} \leq \chi_{6,0.995}^2 = 18.55, \quad (8)$$

after which the error state is injected and reset in the usual way [32]. Two physically motivated bounds close the estimator. After each update the velocity state is clamped isotropically to the maximum credible dock speed, $\mathbf{v} \leftarrow \mathbf{v} \cdot \min(1, v_{\text{max}}/\|\mathbf{v}\|)$ with $v_{\text{max}} = 0.2$ m/s: a measurement implying a faster dock is ego-motion leakage, not dock motion. A covariance ceiling declares the estimate stale once the position uncertainty exceeds 0.15 m (Appendix A.5), so controllers hold rather than act on an unconstrained state. Initialisation requires at least two consensus-surviving markers, since a single planar marker leaves the PnP flip ambiguity unresolved; the first accepted measurement seeds the state with its covariance inflated 100 $\times$, and the initial velocity standard deviation is 0.2 m/s in the sway regime and 0 in the static regime.

The static regime is algebraically a constant-position filter. With $\mathbf{v}(0) = \mathbf{0}$, zero initial velocity covariance and zero velocity process noise, every term that could grow the velocity covariance or its cross-covariance with position is zero, so both remain identically zero; the Kalman gain rows onto the velocity state are therefore zero for every update, and $\mathbf{v} \equiv \mathbf{0}$ for all time. The baseline of Part A is this filter, and the feedforward it feeds is exactly zero by construction.

A.5 Health classification

Downstream consumers see only a semantic health signal computed from the age a of the last accepted measurement and the position uncertainty $\sigma_p = \max_k \sqrt{\mathbf{P}_{kk}}$ over the position block: WARMING before initialisation; STALE if $a > 3$ s or $\sigma_p > 0.15$ m; HEALTHY if $a \leq 0.5$ s and $\sigma_p \leq 0.04$ m; DEGRADED otherwise. The supervisory state machine treats WARMING and STALE as perception loss.